

Supplementary Material:

Effect of boron insertion on metal-film wetting of Fe on MgO

Andi Muhammad Nur Fitrah Syamsul*

Kohji Nakamura

September 18, 2026

S1 Performance of the biased exploration in finding the global minimum

The main text uses only structures from iteration ≥ 10 , because relaxation of candidates begins at that iteration. Here we instead use all iterations of the Fe/MgO model—the model with the largest number of independent searches (13)—to characterise how the biased exploration progresses towards the global minimum, and to show what the discarded early iterations contain. Energies are given as $\Delta E/N = (E - E_{\text{globalmin}})/N$ relative to the lowest energy found anywhere in the 13 searches, so the best-known energy descends to zero (Fig. S1).

The onset of relaxation is the pivot of the search

Across iterations 1–9 the best-known energy falls only from 0.494 to 0.435 eV/atom—0.06 eV/atom, about 12 % of the total descent of the run. At iteration 10, the first iteration at which candidates are surrogate-relaxed, it drops from 0.435 to 0.250 eV/atom in a single iteration: approximately half of the entire descent of the run (49 %) occurs at the onset of relaxation. The per-search drop across the onset has a median of 0.165 eV/atom and a range of 0.084–0.233 eV/atom, so the step is a property of every search rather than of one.

Before the onset the candidates cannot be ranked

The structures evaluated in iterations 1–9 are unrelaxed placements, so their energy ordering reflects the starting geometry rather than any relaxed configuration: several searches plateau after the second or third iteration and show no further improvement for the remainder of the pre-relaxation phase. This is why the main-text analysis discards them, and why their exclusion is not a loss of evidence.

After the onset the descent proceeds in a long, progressively finer sequence

The best-known energy reaches 0.219 ($i = 20$), 0.078 ($i = 30$) and 0.030 eV/atom ($i = 50$)—84 % of the total descent complete by iteration 30 and 94 % by iteration 50—with successive crossings of 0.20 eV/atom at $i = 23$, 0.10 at $i = 29$, 0.05 at $i = 46$, 0.02 at $i = 57$ and 0.005 at $i = 72$. The global minimum is first reached at iteration 77 by one of the searches, and the searches are still making small improvements when the run ends at iteration 100.

The searches do not agree on the answer

Only 10 of the 13 searches end within 0.05 eV/atom of the global minimum, and only 4 within 0.02; the median search finishes 0.040 eV/atom above it (per-search final values 0.000–0.086 eV/atom). A single search supplies the global minimum and a second comes within 0.001 eV/atom of it. The final energy reached by a

*Graduate School of Engineering, Mie University, Tsu, Mie 514-8507, Japan.

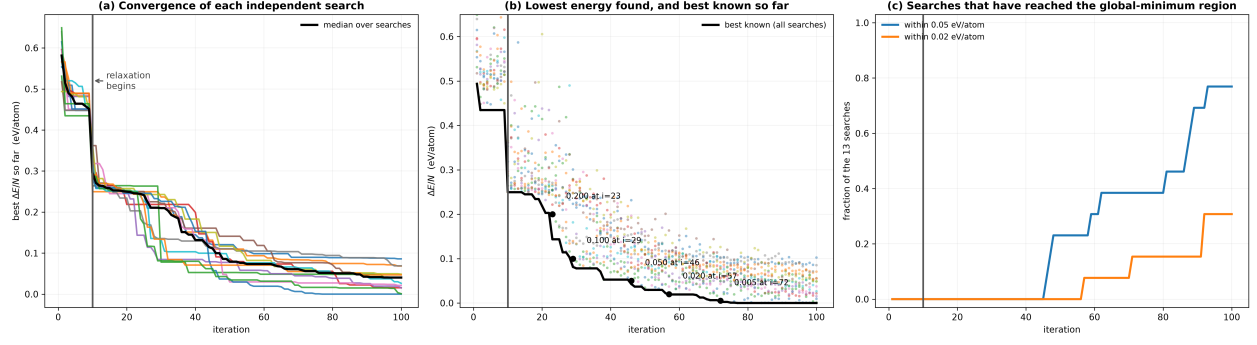


Figure S1: Best-so-far $\Delta E/N$ against iteration for each of the 13 searches, with the median over searches in black; (b) the lowest energy found in each iteration (points) and the best-known energy across all searches (black line), with the iterations at which it first crosses 0.20, 0.10, 0.05, 0.02 and 0.005 eV/atom marked; (c) the fraction of searches whose own best-so-far has come within 0.05 and 0.02 eV/atom of the global minimum. The vertical line in each panel marks iteration 10, where relaxation begins.

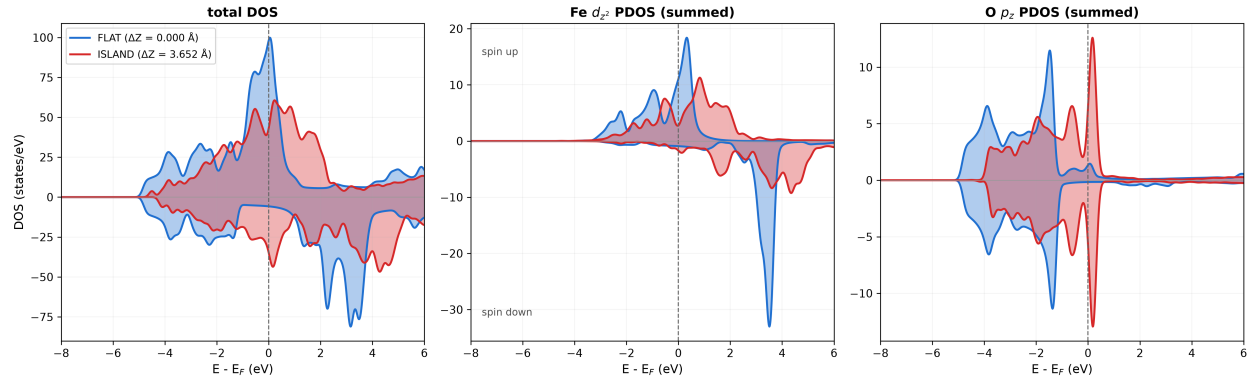


Figure S2: Total DOS, Fe- d_{z^2} and O- p_z for the flat monolayer (blue) and the island ground state (red).

search therefore carries a sizeable spread, which is why the main text compares basins across searches rather than quoting one structure per model.

S2 Electronic-structure origin of the flat $\rightarrow$ island transition

Section IV A of the main text attributes the island ground state to the loss of forced interfacial coupling. Here we test that reading directly, by comparing the flat reference monolayer ($\Delta Z = 0.000 \text{ \AA}$) with the island ground state ($\Delta Z = 3.652 \text{ \AA}$) of the Fe/MgO model in the *same* projection set, so the two are directly comparable (GPAW LCAO/dzp, PBE, kpts (12, 12, 1), 2000 points, width 0.15 eV, Fermi-shifted). Projections are per atom: Fe- d_{z^2} ($l = 2, m = 2$) and O- p_z ($l = 1, m = 0$). d -band moments are taken over $[-5, +3]$ eV around E_F . All numbers in this section are those of Table S1 and Table S2.

Islanding reduces Fe–O hybridisation

[SI-1] Going flat $\rightarrow$ island, the Fe d -band centre rises by +0.83 eV while the O- p_z integral falls by 3.4 % and the Fe- d_{z^2} manifold becomes slightly narrower (−0.10 eV) and more filled (+2.16). The flat monolayer is therefore the more strongly Fe–O hybridised of the two: it mixes more O- p character and its Fe d -states sit lower, whereas the island’s d -states shift up and localise.

Table S1: Density-of-states metrics for the two configurations. *[SI-1, SI-2]*

Quantity	Flat	Island	Δ (island – flat)
d -band centre (eV)	−0.2294	+0.6013	+0.83 (up)
d -band width (eV)	1.4171	1.3139	−0.10 (narrower)
Fe- d_{z^2} integral	31.7136	33.8763	+2.16 (more filled)
O- p_z integral	43.8799	42.3738	−1.51 (−3.4 %)
Spin polarisation	5.8050	4.3695	−1.44 (less magnetic)
DOS at E_F	103.9621	78.0885	−25.9 (−25 %)

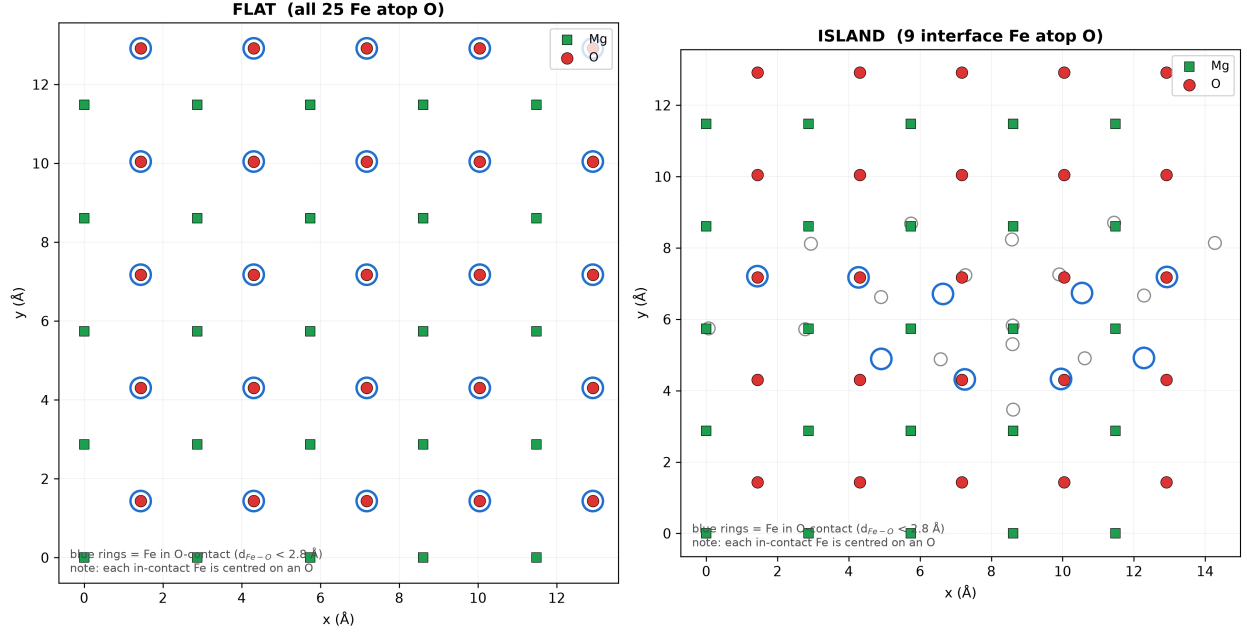


Figure S3: Top view of the interface, showing the metal atoms registered directly above the substrate oxygen on the MgO(001) lattice.

Islanding weakens the magnetic and electronic activity at E_F

[SI-2] The spin polarisation falls from 5.81 to 4.37 and the DOS at the Fermi level drops by 25 % (104.0 $\rightarrow$ 78.1). The flat monolayer is the electronically “hotter” configuration; the island is quieter.

Defining the interface by geometry, not by height

The island is a buckled cluster rather than a stack of layers, so its interface cannot be defined by a height criterion: a z -based layer split would cut through the cluster and misassign atoms as a function of its buckling. We therefore define the interface geometrically: interface Fe := Fe with a nearest O within 2.8 Å, i.e. Fe actually in contact with the oxide. By this criterion the flat monolayer is 25/25 interface atoms and the island only 9/25.

The island’s true interface Fe are not flat-like

[SI-3] Restricting the comparison to the 9 Fe atoms genuinely in O contact does not recover the flat signature: their d -band centre is +0.5106 eV, against −0.2294 eV for the flat monolayer—a difference of 0.74 eV, larger than the spread between the island’s interface (+0.5106) and non-interface (+0.6524) groups. Two features of the model matter here. First, the two groups sit at essentially the same nearest-O distance (2.3311 Å island vs 2.3000 Å flat), and that distance is the one set when the reference film is built—a 2.3 Å interfacial

Table S2: Site-resolved d -band metrics and the Fe-on-O registry, by geometric group. The last two columns are the mean nearest-O distance and the mean in-plane offset from the nearest substrate atom (0 = directly atop); for groups that include non-contacting Fe these are not bond lengths. [*SI-3*, *SI-4*]

Structure	Group	n	d -centre (eV)	width (eV)	f	$\langle \text{nearest } d_{\text{Fe-O}} \rangle$ (Å)	$\langle \text{offset} \rangle$ (Å)
Flat	interface (= all)	25	-0.2294	1.4171	31.7136	2.3000	0.0000
Island	interface ($d_{\text{Fe-O}} < 2.8$ Å)	9	+0.5106	1.5165	12.2134	2.3311	0.3722
Island	non-interface	16	+0.6524	1.1815	21.6629	4.3462	0.4726
Island	all	25	+0.6013	1.3139	33.8763	3.6208	0.4365

separation that relaxation preserves (Sec. II C of the main text)—rather than an emergent quantity, so the electronic difference cannot be a per-bond-length effect. Second, the flat layer’s registry is perfect (offset 0.0000 Å) and uniform, whereas the island’s 9 contacts are buckled (offset 0.3722 Å) and only a third of the film. The flat monolayer’s strong Fe–O coupling is therefore a collective property of all 25 metal atoms being registry-locked in one plane, not a property of an individual Fe–O bond.

The Fe-atop-O registry is a consistency check

[*SI-4*] Every in-contact Fe sits directly atop an oxygen—25/25 in the flat monolayer and 9/9 in the island—and none atop Mg. This is the registry determined experimentally for the first monolayer of Fe on MgO(001) by LEED I–V analysis [1] and used in first-principles models of the Fe|MgO|Fe interface [2], and it is also the registry the reference layer is built with—the construction places the substrate oxygen directly above the metal sites—so the flat film’s registry follows from the construction as much as from the physics. It is reported here as a consistency check, not as a prediction of the search. What the search does add is the island’s behaviour: the Fe that remain in contact keep O as their nearest in-plane neighbour (9/9) rather than switching to Mg, so the island reduces the *number* of Fe–O contacts (25 $\rightarrow$ 9) without abandoning their character.

Combined picture

The flat monolayer maximises Fe–O coupling by locking every metal atom into one registered plane, at the cost of the three-dimensional Fe–Fe coordination available to a cluster; the island reverses that trade, retaining strong individual Fe–O bonds for the third of its atoms that remain in contact while restoring metal cohesion for the rest. This is the mechanism stated in Sec. IV A of the main text as reduced forced interfacial coupling plus restored metal cohesion. It is explicitly not a lattice-strain effect: the in-plane mismatch of this interface is carried by the substrate, which is held fixed, and the metal film sits at its own equilibrium lattice constant (Sec. II A and IV D of the main text).

S3 Method-parameter sensitivity of the biased exploration

The search used here is a custom, biased scheme—GOFEE (a GPR surrogate with an LCB acquisition function) seeded from a flat Fe reference layer, with a heterostructure-aware randomiser and a rattle generator supplying the perturbation. Here we vary three method parameters on the same $\text{Fe}_{25}\text{Mg}_{25}\text{O}_{25}$ system and ask whether the method’s conclusions survive the choice: is the island still the ground state, how far does a single search reach, and is the flat/island basin picture preserved. The baseline for every family is the plain Fe/MgO run (rattle 1.5/2.3, $\kappa = 2$, no dipole correction). Energies are per-seed best $\Delta E/N$ relative to the lowest energy found in any of these runs, -436.909 eV.

Only completed searches are used

Several of the searches in this study were stopped early, and they are excluded everywhere in this paper. A short search has had less time to descend, so its per-seed best is systematically worse—every such search in this set is a severe outlier (0.23–0.35 eV/atom, against ~ 0.03 –0.09 for completed searches)—and because searches that stopped early are not distributed evenly across the settings they would distort the comparison

Table S3: Method-parameter sensitivity; completed searches only (100 iterations). *[SI-5, SI-6, SI-7]*

Family	Setting	Searches	per-seed best (eV/atom)	Flat fraction	Diversity
rattle	baseline (1.5 / 2.3)	13	0.03683 $\pm$ 0.02575	0.165	2.22
rattle	reduced-0.5 (1.0 / 1.8)	2	0.26073 $\pm$ 0.06804	0.023	3.74
rattle	reduced-1.0 (0.5 / 1.3)	4	0.25583 $\pm$ 0.13287	0.017	4.33
kappa	$\kappa = 2$ (baseline)	13	0.03683 $\pm$ 0.02575	0.165	2.22
kappa	$\kappa = 1$	16	0.03947 $\pm$ 0.02371	0.182	2.36
kappa	$\kappa = 3$	10	0.03509 $\pm$ 0.02182	0.149	2.65
kappa	$\kappa = 4$	9	0.03209 $\pm$ 0.01872	0.132	2.68
dipole	no dipole (baseline)	13	0.03683 $\pm$ 0.02575	0.165	2.22
dipole	dipole xy	11	0.03770 $\pm$ 0.02436	0.158	2.67

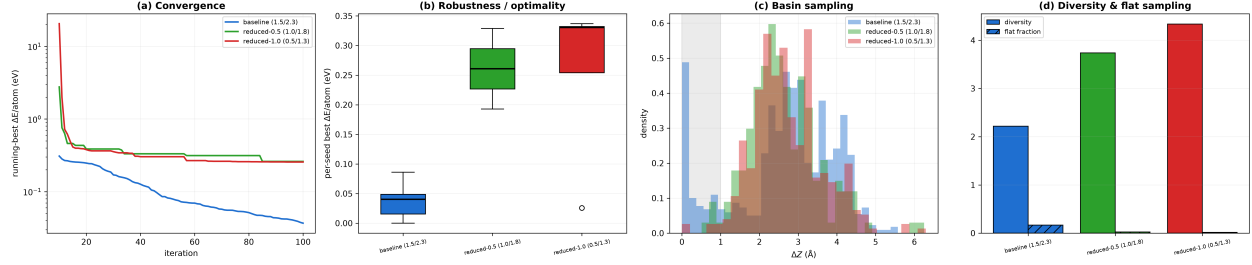


Figure S4: Convergence, per-seed best, basin sampling and diversity for the rattle-strength family (four panels).

between them. The same criterion is applied without exception throughout the analysis, and it is decided from the iteration count recorded for each search, not from how a search is labelled: a search that looks complete can have stopped early. The baseline quoted in Table S3 therefore consists of 13 completed searches—one further search of that family stopped early—which is the replica count used in the main text.

Reducing the rattle strength degrades the search by about $7\times$

[SI-7] Cutting the two perturbation amplitudes roughly in half raises the per-seed best from 0.0368 to 0.2607 / 0.2558 eV/atom—a factor of 7.1 and 7.0—and the flat-basin fraction collapses from 0.165 to 0.023 / 0.017. Reducing the perturbation is therefore not a cheaper equivalent of the baseline: the rattle is what makes the search escape the initial region and reach the low-energy basin at all, and less of it leaves the search sampling almost exclusively the island side. The direction is not in doubt: every arm of the sweep is worse than the baseline, and the flat basin is under-sampled by an order of magnitude in the flat fraction.

The kappa parameter changes nothing measurable

[SI-5] All four settings land within 0.0321–0.0395 eV/atom—a total span of 0.0074 eV/atom, far inside the standard deviations (0.019–0.026)—so the method’s conclusions are robust to the acquisition parameter, and in particular no value of κ can be identified as better than another from these runs. There is a small, monotone trend in the flat-basin fraction—0.182, 0.165, 0.149 and 0.132 for $\kappa = 1, 2, 3$ and 4—i.e. more exploration relative to exploitation samples slightly less of the flat basin, but the effect is within the same order as the seed-to-seed spread.

The dipole correction does not change the outcome

[SI-6] The per-seed best moves from 0.03683 $\pm$ 0.02575 to 0.03770 $\pm$ 0.02436 eV/atom, a difference of 0.0009 eV/atom—an order of magnitude smaller than the standard deviation.

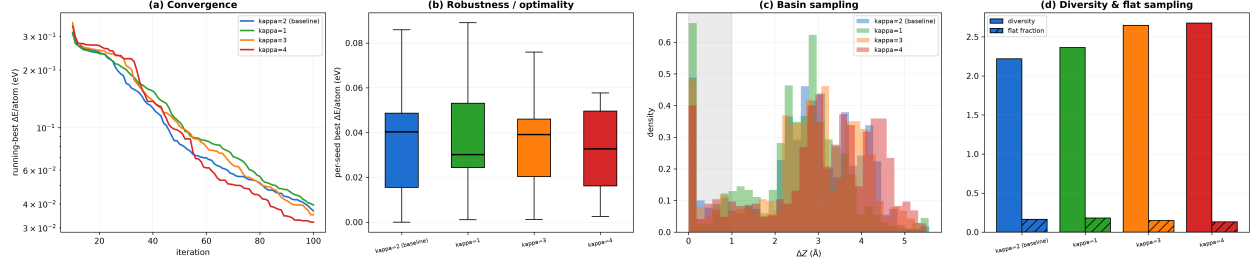


Figure S5: The same four panels for the kappa family.

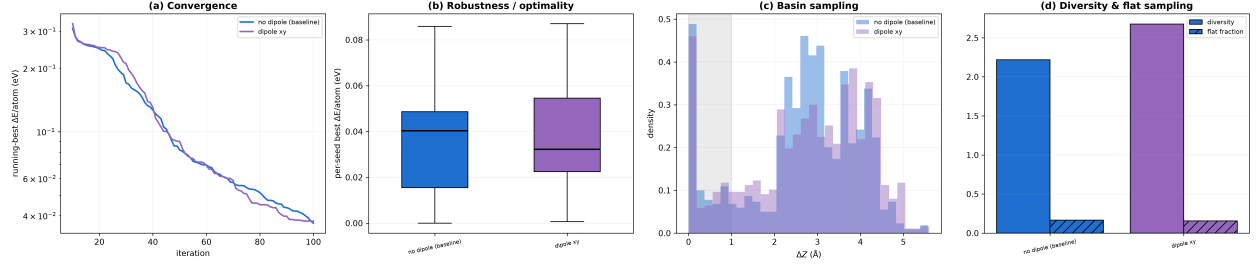


Figure S6: The same four panels for the dipole family.

S4 The result does not depend on the search budget

The main text reports the Fe/MgO model at a 100-iteration budget. To test whether the *comparison* the paper makes—the island being the ground state and the flat basin sitting above it—survives a longer search, the same calculation was run to 200, 400 and 600 iterations: the run script is identical to the main-text one except for the number of iterations, and the reference layer, the two perturbation generators and the acquisition parameter are unchanged. 6, 7 and 4 searches completed at 200, 400 and 600 iterations respectively.

Method

The primary evidence is a per-run truncation: each search is cut at a budget k (100, then its own full budget) and the flat/island quantities recomputed, so the 100-iteration and full-budget numbers come from the *same trajectory* and need no cross-run normalisation. The flat/island split is $\Delta Z \leq 1.0$ Å and only iterations ≥ 10 are used, as in the main text. Each arm is tested against its own budget, and searches that stopped early are excluded.

The island remains the ground state at every budget

In all 17 searches the lowest-energy structure is an island at both the 100-iteration cut and the full budget. (*MT-1*)

The flat–island separation does not shrink with a longer search

Per run, the separation is unchanged in 5 of the 17 searches and larger in the other 12—it never decreases—by a median of +0.0104 eV/atom (range 0 to +0.0268). Pooled per arm it rises from 0.1901 to 0.1901 (200 iterations), 0.1624 to 0.1662 (400) and 0.1807 to 0.1978 eV/atom (600). A longer search therefore does not bring the flat basin closer to the ground state; if anything it leaves it relatively higher. (*SI-9*)

The flat basin is a smaller fraction of the sampled set at longer budgets (0.165 at 100 iterations falling to 0.05–0.11 at 600). Longer searches keep finding better islands, so the flat basin occupies a smaller share of the sampled structures.

Table S4: Flat-basin minimum $\Delta E/N$ (eV/atom), pooled per arm at each budget (the main-text definition: relative to the arm’s own global minimum). The main-text model is the 100-iteration reference.

Budget	completed searches	$k = 100$	$k = 200$	$k = 400$	$k = 600$
200 iterations	6	0.1901	0.1901	—	—
400 iterations	7	0.1624	0.1645	0.1662	—
600 iterations	4	0.1807	0.1932	0.1966	0.1978
main text (13 searches, 100 it)	13	0.1888			

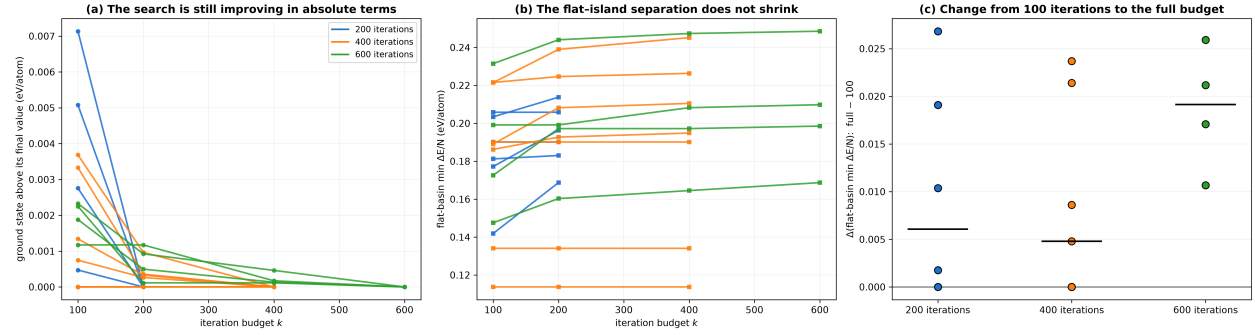


Figure S7: (a) the absolute ground-state energy of each search against the budget, showing that the search is still improving in absolute terms; (b) the flat-island separation of each search against the budget, showing that it does not shrink; (c) the per-search change in the flat-basin minimum from 100 iterations to the full budget.

Cross-check

The 13 main-text searches, read through this analysis, reproduce the pooled flat-basin minimum of 0.1888 eV/atom exactly, so the two analyses agree on the reference value.

S5 The result does not depend on which phase sets the in-plane lattice

The main-text model takes the Fe lattice constant for the simulation cell, so the substrate carries the in-plane mismatch (compressed 3.6% against its bulk value) and the film is unstrained—the inverse of the experimental stack, where bulk MgO imposes its lattice on a thin film. To test whether the structural result depends on this choice, the cell is swept from Fe-matched to MgO-matched, $a = a_{\text{Fe}} + f \cdot (a_{\text{MgO}}/\sqrt{2} - a_{\text{Fe}})$, with both the film and the substrate built on that cell. The two endpoints are the experimental lattice constants of the two bulk phases: bcc α -Fe, $a = 2.866 \text{ \AA}$ [3], and rocksalt MgO, $a = 4.2112 \text{ \AA}$ [4] (4.212 $\text{\AA}$ as used in the inputs). At $f = 0.25$ the substrate is compressed 2.83% and the film stretched 0.98%; at $f = 0.75$, -0.94% and $+2.94\%$; at $f = 1.00$ the substrate sits at the experimental MgO lattice constant and the film is stretched 3.92%—the convention of the experimental stack. 10, 10 and 5 searches completed at $f = 0.25$, 0.75 and 1.00. The paper’s own Fe/MgO model (13 searches, cell = a_{Fe}) is the Fe-matched reference; it uses the *computed* Fe lattice constant (2.87019 $\text{\AA}$) rather than the experimental 2.866 $\text{\AA}$, so the sweep’s $f = 0$ endpoint is a reference, not identical to the main-text model.

The island is the ground state under every constraint

In all 25 searches, at every value of f , the lowest-energy structure is an island. (*MT-1*)

Table S5: The constraint, the strains it puts on each phase, and the flat-basin minimum (pooled, relative to each arm’s own global minimum).

f	cell a (Å)	substrate strain	film strain	searches	flat-basin min $\Delta E/N$ (eV/atom)	island ground state
0 (main text)	2.87019	−3.63 %	0.00 %	13	0.1888	yes
0.25	2.89408	−2.83 %	+0.98 %	10	0.1932	yes
0.75	2.95025	−0.94 %	+2.94 %	10	0.1982	yes
1.00	2.97833	0.00 %	+3.92 %	5	0.1907	yes

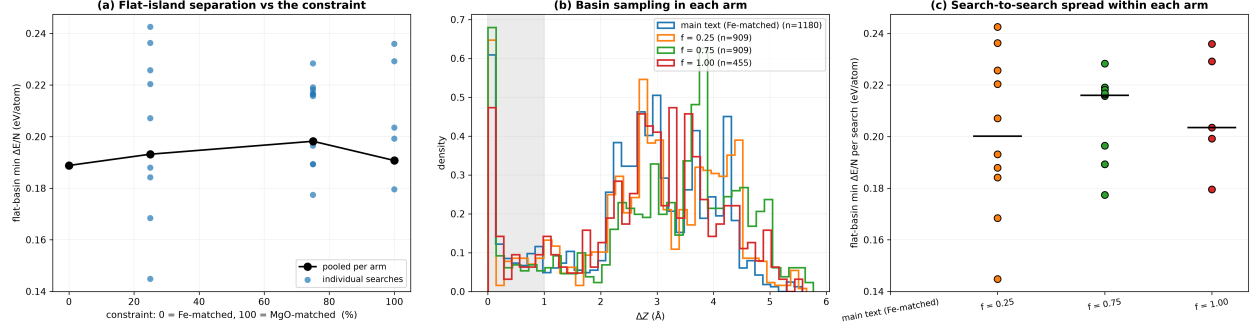


Figure S8: (a) the flat-basin minimum against the constraint, pooled per arm with the individual searches overlaid; (b) the ΔZ distributions of each arm; (c) the search-to-search spread within each arm.

The flat-basin minimum is flat against the constraint

Pooled, it is 0.1932 ($f = 0.25$), 0.1982 ($f = 0.75$) and 0.1907 eV/atom ($f = 1.00$), against 0.1888 eV/atom for the Fe-matched main-text model—a total spread of 0.0094 eV/atom across the whole sweep. The search-to-search scatter within an arm is 0.017–0.031 eV/atom, i.e. 2–3 $\times$ the entire sweep effect, so no dependence on the constraint is resolvable. For scale, the constraint moves the separation by about a quarter of the boron effect (0.040 eV/atom). (*SI-10*)

The physically inverted strain case is therefore calculated for this model

The far end of the sweep places the substrate at the experimental MgO lattice constant and stretches the film—the convention of the experimental stack—and the result is unchanged.

S6 The inverted stack: a ground-state comparison (an MgO film on an Fe substrate)

The main text studies Fe deposited on MgO, and its ground state is an island. Here we ask whether the wetting of the deposited layer is symmetric—whether a film behaves the same way when it is the oxide that is deposited. The comparison is made at the ground state of each stack: an MgO film on an Fe substrate, against the Fe-on-MgO result of the main text. The inverted stack uses the same system and scheme ($\text{Fe}_{25}\text{Mg}_{25}\text{O}_{25}$, 100 iterations, $\kappa = 2$, the same two perturbation generators, cell at the experimental Fe lattice constant, 2.866 Å).

Method

The flatness is measured over the deposited film in each stack—Fe for Fe-on-MgO, MgO (Mg + O) for MgO-on-Fe—so the comparison is apples-to-apples on wetting. (ΔZ over Fe in the inverted stack would measure the substrate, which is flat by construction.) The ground state is the lowest-energy structure of a stack, and the flat/island split is $\Delta Z \leq 1.0$ Å of the deposited film. Only iterations ≥ 10 are used, and only completed searches enter the comparison.

Table S6: The two stacks at their ground states.

Stack	Ground state	ΔZ of the deposited film	Flat film relative to the island
Fe on MgO (main text)	island	3.65 Å	+0.189 eV/atom above
MgO on Fe (inverted)	flat film	0.39 Å	-0.016 eV/atom below

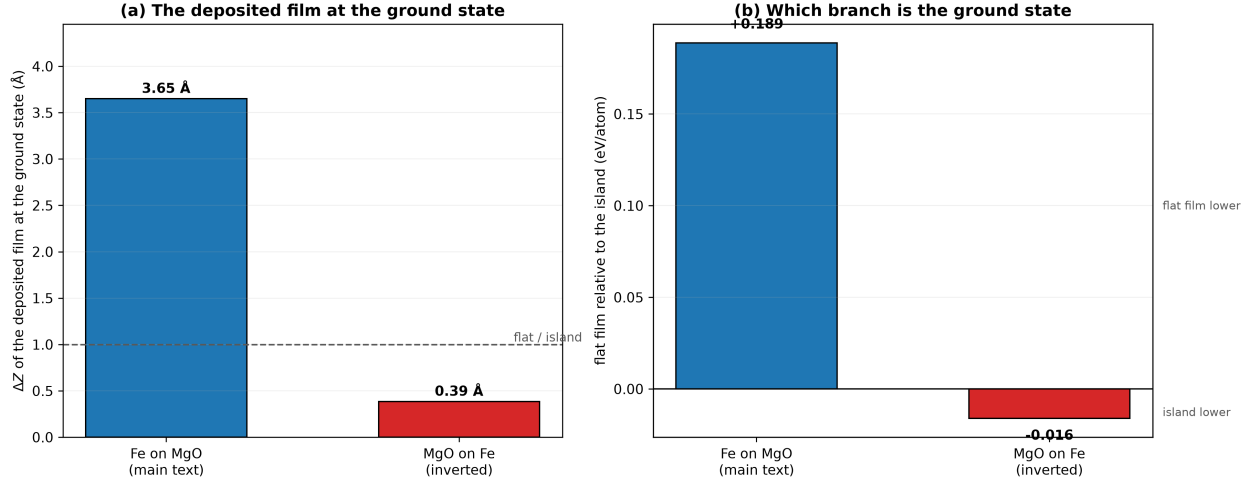


Figure S9: (a) ΔZ of the deposited film at the ground state of each stack, against the 1.0 Å flat/island threshold; (b) where the flat film lies relative to the island for each stack—positive means the island is the ground state, negative means the flat film is.

The ground state of the inverted stack is a flat MgO film—the opposite of Fe-on-MgO

On the inverted stack the lowest structure found is a flat MgO film ($\Delta Z = 0.39$ Å), and the flat film is 0.016 eV/atom below the corresponding island. For Fe-on-MgO the ordering is the reverse: the island is the ground state ($\Delta Z = 3.65$ Å), and the flat film sits 0.189 eV/atom above it. The sign of the wetting preference therefore inverts between the two stacks—the layer that is deposited keeps its flat, wetting configuration when it is the oxide, and breaks up into an island when it is the metal. (SI-11)

References

- [1] Toshio Urano and Toru Kanaji. Atomic and electronic structure of ultrathin iron film on MgO(001) surface. *Journal of the Physical Society of Japan*, 57(10):3403–3410, 1988. doi: 10.1143/JPSJ.57.3403.
- [2] W. H. Butler, X.-G. Zhang, T. C. Schulthess, and J. M. MacLaren. Spin-dependent tunneling conductance of Fe|MgO|Fe sandwiches. *Physical Review B*, 63(5):054416, 2001. doi: 10.1103/PhysRevB.63.054416.
- [3] Paul Pietrokowsky. Lattice parameter of alpha iron by divergent beam diffraction. *Journal of Applied Physics*, 37(12):4560–4571, 1966. doi: 10.1063/1.1708081.
- [4] Howard E. Swanson and Eleanor Tatge. Standard x-ray diffraction powder patterns. National Bureau of Standards Circular 539, Vol. 1, 1953.